الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 9 · LESSONS 3-3 · 3-4 · 3-5

Weekly Quiz Preparation

Grade 10 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

Mr. Yousef Al Yousef · **Friday 30 October 2026**

1 AA Similarity

التشابه بالزاويتين AA

AA Similarity Postulate: if two angles of one triangle are congruent to two angles of another, the triangles are similar. Use $180°$ – the two known angles to get the third angle.

مسلّمة AA: إذا تطابقت زاويتان في مثلث مع زاويتين في مثلث آخر فالمثلثان متشابهان. أوجد الزاوية الثالثة بـ $180°$ − الزاويتين.

Steps الخطوات

1	**Find the third angle of each triangle: $180°$ – (the two known)**	أوجد الزاوية الثالثة
2	**List the angles of each triangle**	اكتب زوايا كل مثلث
3	**Two pairs of congruent angles? → similar by AA**	زوجان متطابقان ← تشابه AA
4	**Write the similarity statement with the vertices in matching order**	اكتب عبارة التشابه بالترتيب

Worked example مثال محلول

$\triangle ABC$ has $m\angle A = 65°$ and $m\angle B = 50°$; $\triangle DEF$ has $m\angle D = 65°$ and $m\angle E = 65°$. Are the triangles similar? Explain.

$m\angle C = 180 - 65 - 50 = 65°$ — the angles of a triangle add to 180° / مجموع زوايا المثلث $180°$

$m\angle F = 180 - 65 - 65 = 50°$ — the same for the second triangle / الأمر نفسه للمثلث الثاني

match equal measures / طابِق الزوايا ذات القياس نفسه

$\angle A \cong \angle D\ (65°),\quad \angle B \cong \angle F\ (50°)$

AA Similarity Postulate / مسلّمة التشابه AA

two pairs of congruent angles: $\triangle ABC \sim \triangle DFE$ (AA)

Watch out انتبه

- Only two matching pairs are needed, but you must show them. — يكفي زوجان متطابقان لكن يجب أن تُظهرهما.
- Match angles by their MEASURES, not by their letters. — طابِق الزوايا بقياساتها لا بحروفها.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. $\triangle STR$ has $m\angle S = 35°$ and $m\angle T = 85°$; $\triangle UVW$ has $m\angle U = 35°$ and $m\angle V = 60°$. Are the triangles similar? Explain.
2. One triangle has angles $70°$ and $40°$; another has angles $70°$ and $50°$. Are the triangles similar? Explain.
3. Two right triangles each have an acute angle of $38°$. Are they similar? Explain.

1 AA Similarity

التشابه بالزاويتين AA

Lesson 3-3 · Part 1 · p. 133

MAT.3.05.04.005

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 12 | MCQ · on the computer | اختيار من متعدد · على الحاسوب | Exercises 1-8 · p. 133

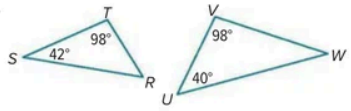
Example 1

Determine whether each pair of triangles is similar. Explain your reasoning.

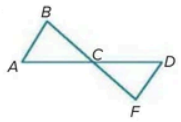
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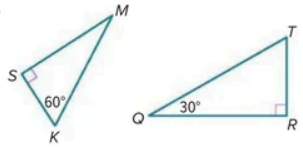
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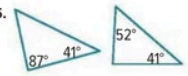
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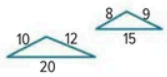
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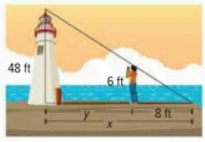
Example 2

7. CELL TOWERS A cell phone tower casts a shadow that is 100 feet long. At the same time, Lia stands near the tower and casts a shadow that is 3 feet 4 inches long. If Lia is 4 feet 6 inches tall, how tall is the cell phone tower?

8. LIGHTHOUSE Maya wants to know how far she is standing from a lighthouse. The end of Maya's shadow coincides with the end of the lighthouse's shadow.

a. What is the distance from the lighthouse to the end of the lighthouse's shadow, x?

b. What is the distance from Maya to the lighthouse, y?



2 Similar triangles in problems
المثلثات المتشابهة في المسائل

Similar triangles have proportional corresponding sides. Name the triangles, write the proportion with corresponding sides in the same order, cross multiply, solve.

المثلثات المتشابهة أضلاعها المتناظرة متناسبة. سمِّ المثلثين، اكتب النسبة بالترتيب نفسه، اضرب تبادليًا وحلّ.

Steps الخطوات

1	**Name the similar triangles and the criterion (AA)**	سمِّ المثلثين والمعيار
2	**Write every length in the same unit**	وحّد الوحدات
3	**Write corresponding sides in the same order as a proportion**	اكتب النسبة بالترتيب نفسه
4	**Cross multiply, solve, give the unit**	اضرب تبادليًا وحلّ واذكر الوحدة

Worked example مثال محلول

A tree casts a shadow 12 ft long. At the same time a person 5 ft tall casts a shadow 4 ft long. How tall is the tree?

the triangles are similar (AA): the same sun angle, right angles
tree and shadow, perso
الشخص وظله

$$\frac{h}{5} = \frac{12}{4}$$

height ÷ height = shadow ÷ shadow (same units)
الارتفاع ÷ الارتفاع = الظل ÷ الظل (بوحدة واحدة)

$$h = \frac{5 \cdot 12}{4} = 15 \text{ ft}$$

cross multiply and divide
اضرب تبادليًا ثم اقسم

Watch out انتبه

- 3 ft 4 in is 40 inches, not 3.4.
- Height goes with height, shadow with shadow.

3 أقدام و4 بوصات = 40 بوصة وليس 3.4.
الارتفاع مع الارتفاع والظل مع الظل.

Now you try جرّب بنفسك solutions at the back الحلول في آخر الكتيّب

1. A flagpole casts a shadow 18 ft long. At the same time a person 6 ft tall casts a shadow 4 ft long. How tall is the flagpole?
2. A building casts a shadow 30 ft long. At the same time a person 5 ft 3 in. tall casts a shadow 42 in. long. How tall is the building?
3. Right angles at H and K; $IH = 6$, $HJ = 9$, $JK = x$, $KL = 4$; $\overline{HK}$ and $\overline{IL}$ cross at J. Identify the similar triangles and find x.

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

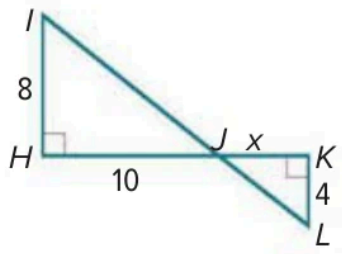
من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 23 · Written · on paper · كتابي · على الورق · Exercises 5-11 · p. 139

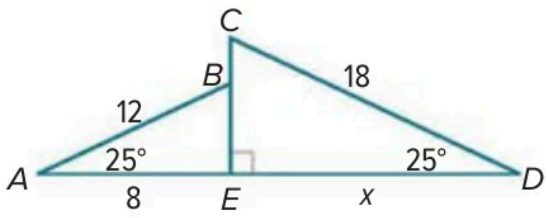
Example 2

Identify the similar triangles. Then find the value of *x*.

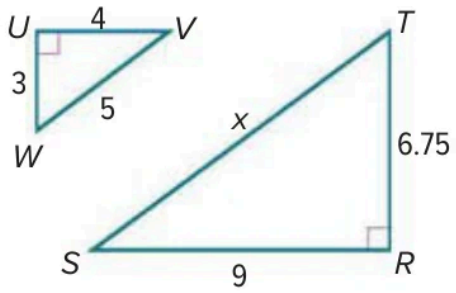
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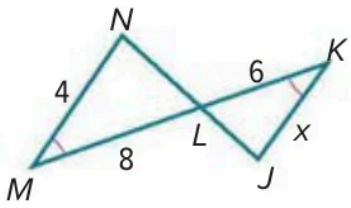
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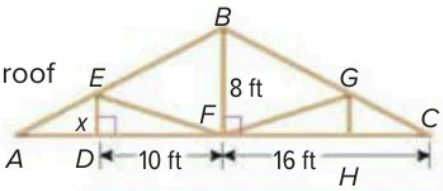


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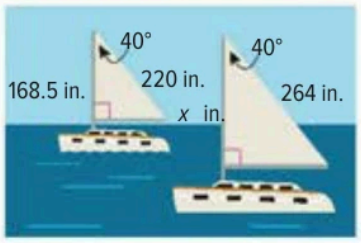
Example 3

9. ROOFING The skeleton of a roof is shown. Find the value of *x* such that triangles *DEF* and *FBC* in the outline of the roof are similar



10. RADIO A radio tower casts an 8-foot-long shadow at the same time that a vertical yardstick casts a shadow one half inch long. If the triangles formed by the objects and their shadows are similar, how tall is the radio tower?

11. SAILING The two sailboats shown are participating in a regatta. If the sails are similar, what is the value of *x*?



3 Midsegments and parallel lines
القطعة المتوسطة والمستقيمات المتوازية

Triangle Midsegment Theorem: the segment joining the midpoints of two sides is parallel to the third side and half its length. Congruent parts cut by parallel lines are equal: set them equal.

القطعة المتوسطة توازي الضلع الثالث وطولها نصفه. والأجزاء المتطابقة التي تقطعها مستقيمات متوازية متساوية: ساوِ بينها.

Steps الخطوات

1	**Midsegment: its length = ½ the side it is parallel to**	طول المتوسطة = نصف الضلع الموازي
2	**Side $= 2 \times$ midsegment**	الضلع = ضعف المتوسطة
3	**Parallel lines with congruent parts: set the two expressions equal**	ساوِ بين تعبيري الجزأين
4	**Solve and check**	حلّ وتحقّق

Worked example مثال محلول

$\overline{JH}$ is a midsegment of $\triangle KLM$, parallel to $\overline{LM}$ with $LM = 24$. Find JH. Then, parallel lines cut congruent parts $x + 4$ and $2x - 1$: find x.

$JH = \frac{1}{2}LM = \frac{1}{2}(24) = 12$ — the Triangle Midsegment Theorem / نظرية القطعة المتوسطة في المثلث

$x + 4 = 2x - 1$ — congruent parts are equal / الأجزاء المتطابقة متساوية

$5 = x \Rightarrow x = 5$ — subtract x, add 1 / اطرح x وأضف 1

Watch out انتبه

- Halve the SIDE to get the midsegment; double the midsegment to get the side.
- Set equal only the parts marked congruent.

نصّف الضلع لتحصل على المتوسطة، وضاعف المتوسطة لتحصل على الضلع.
ساوِ فقط بين الأجزاء المعلَّمة بأنها متطابقة.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. The midsegment of a triangle is 9 cm long. How long is the side it is parallel to?
2. $\overline{DE}$ is a midsegment of $\triangle ABC$, parallel to $\overline{BC}$, with $DE = 2x + 3$ and $BC = 5x - 6$. Find x.
3. Parallel lines cut congruent parts $3x - 2$ and $x + 8$ on a transversal. Find x.

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

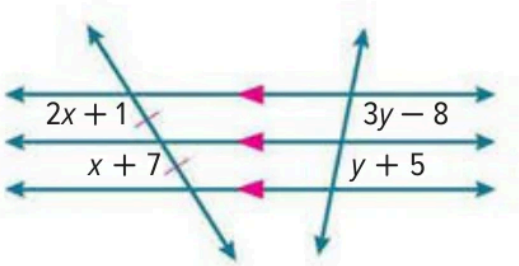
Question 13 | MCQ · on the computer | اختيار من متعدد · على الحاسوب

Exercises 11-16 · p. 146

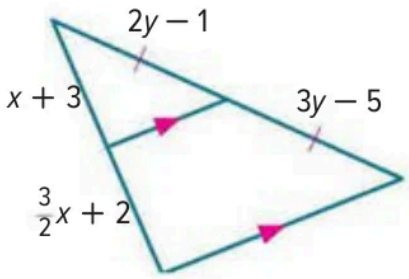
Example 5

Find the values of *x* and *y*.

11.



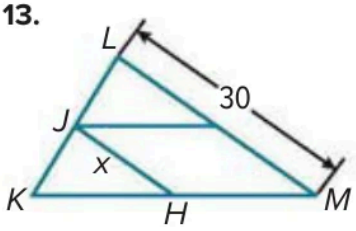
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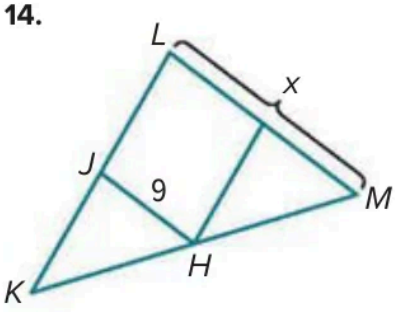
Mixed Exercises

$\overline{JH}$ is a midsegment of $\triangle KLM$. Find the value of *x*.

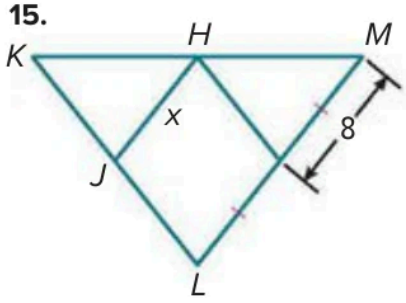
13.



14.



15.



16. In $\triangle ABC$, $\overline{DE}$ is parallel to $\overline{AC}$ and $DE = 10$. Find the length of $\overline{AC}$ if $\overline{DE}$ is a midsegment of $\triangle ABC$.



Solutions

written out step by step · الحلول

1 · AA Similarity — التشابه بالزاويتين AA

1.1

$\triangle STR$ has $m\angle S = 35°$ and $m\angle T = 85°$; $\triangle UVW$ has $m\angle U = 35°$ and $m\angle V = 60°$. Are the triangles similar? Explain.

$m\angle R = 180 - 35 - 85 = 60°$, $m\angle W = 180 - 35 - 60 = 85°$

the third angles — الزاويتان الثالثتان

two pairs of congruent angles — زوجان من الزوايا المتطابقة

$\angle S \cong \angle U$ $(35°)$, $\angle T \cong \angle W$ $(85°)$

similar by AA

~~AA Similarity Postulate~~ — مسلّمة التشابه AA

1.2

One triangle has angles $70°$ and $40°$; another has angles $70°$ and $50°$. Are the triangles similar? Explain.

the third angles — الزاويتان الثالثتان

third angles: $180 - 70 - 40 = 70°$ and $180 - 70 - 50 = 60°$

only the $70°$ angles match — زاويتا $70°$ فقط متطابقتان

angles $70, 40, 70$ and $70, 50, 60$: only one pair matches

not similar

~~AA needs two pairs~~ — تشابه AA يحتاج زوجين

1.3

Two right triangles each have an acute angle of $38°$. Are they similar? Explain.

right angles are congruent — الزوايا القائمة متطابقة

each has a right angle $(90°)$ and a $38°$ angle

two pairs of congruent angles: similar by AA

~~AA Similarity Postulate~~ — مسلّمة التشابه AA

2 · Similar triangles in problems — المثلثات المتشابهة في المسائل

2.1

A flagpole casts a shadow 18 ft long. At the same time a person 6 ft tall casts a shadow 4 ft long. How tall is the flagpole?

$\frac{h}{6} = \frac{18}{4}$

height ÷ height = shadow ÷ shadow — الارتفاع ÷ الارتفاع = الظل ÷ الظل

$h = \frac{6 \cdot 18}{4} = 27$ ft

cross multiply and divide — اضرب تبادليًا ثم اقسم

2.2

A building casts a shadow 30 ft long. At the same time a person 5 ft 3 in. tall casts a shadow 42 in. long. How tall is the building?

$5\text{ ft }3\text{ in.} = 63\text{ in.}, \quad 30\text{ ft} = 360\text{ in.}$ — the same unit / الوحدة نفسها

$\frac{h}{63} = \frac{360}{42}$ — height ÷ height = shadow ÷ shadow / الارتفاع ÷ الارتفاع = الظل ÷ الظل

$h = \frac{63 \cdot 360}{42} = 540\text{ in.} = 45\text{ ft}$ — divide by 12 for feet / اقسم على 12 لتحصل على الأقدام

2.3

Right angles at H and K; $IH = 6$, $HJ = 9$, $JK = x$, $KL = 4$; $\overline{HK}$ and $\overline{IL}$ cross at J. Identify the similar triangles and find x.

right angles and vertical angles / زاويتان قائمتان وزاويتان متقابلتان بالرأس

$\angle H \cong \angle K = 90^\circ,\ \angle IJH \cong \angle LJK:\ \triangle IHJ \sim \triangle LKJ\ (AA)$

$\frac{IH}{LK} = \frac{HJ}{KJ} \Rightarrow \frac{6}{4} = \frac{9}{x}$ — corresponding sides / الأضلاع المتناظرة

$6x = 36 \Rightarrow x = 6$ — cross multiply / اضرب تبادليًا

3 · Midsegments and parallel lines — القطعة المتوسطة والمستقيمات المتوازية

3.1

The midsegment of a triangle is 9 cm long. How long is the side it is parallel to?

$\text{side} = 2 \cdot 9 = 18\text{ cm}$ — the side is twice the midsegment / الضلع ضعف القطعة المتوسطة

3.2

$\overline{DE}$ is a midsegment of $\triangle ABC$, parallel to $\overline{BC}$, with $DE = 2x + 3$ and $BC = 5x - 6$. Find x.

$BC = 2\,DE:\ 5x - 6 = 2(2x + 3)$ — the Midsegment Theorem / نظرية القطعة المتوسطة

$5x - 6 = 4x + 6 \Rightarrow x = 12$ — subtract $4x$, add 6 / اطرح $4x$ وأضف 6

3.3

Parallel lines cut congruent parts $3x - 2$ and $x + 8$ on a transversal. Find x.

$3x - 2 = x + 8$ — congruent parts are equal / الأجزاء المتطابقة متساوية

$2x = 10 \Rightarrow x = 5$ — subtract x, add 2, divide by 2 / اطرح x وأضف 2 ثم اقسم على 2